University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

# STA 2101: Statistics & Probability

## Lecture 11:
## Venn Diagram – Prob Intersection & Union


Atanu Shuvam Roy

Lecturer,
Dept. of CSE, ULAB, Dhaka

EMAIL:
atanu.shuvam@ulab.edu.bd


Class Code
sdb2r4mu

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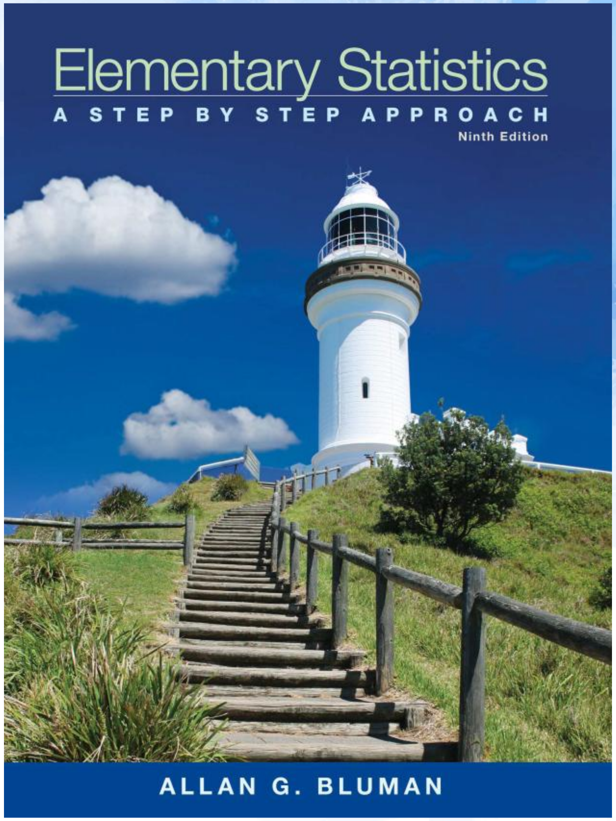
# Reference Book

- **Text Book(s)**:
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

- **Reference Material/Book(s)**:
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross

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# Mark Distribution


| Assessments | % |
|---|---|
| Mid Term Examination | 25 |
| Final Examination | 40 |
| Attendance and Class Participation | 5 |
| CT | 5 |
| Project | 25 |
| Total | 100 |

# Today’s Topics

- Measures of Central Tendency
- Mean
- Median

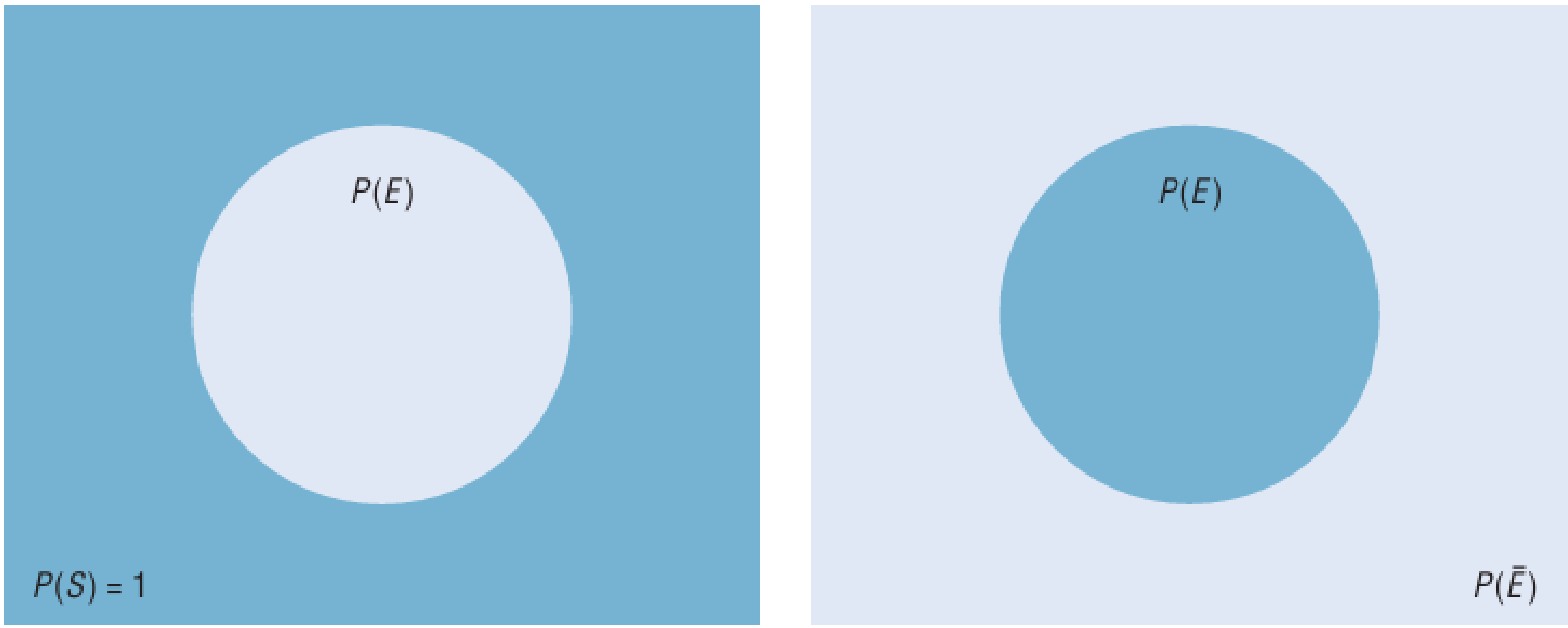
"Probability Of" "Given"

$$P(A \text{ and } B) = P(A) \times P(B \mid A)$$

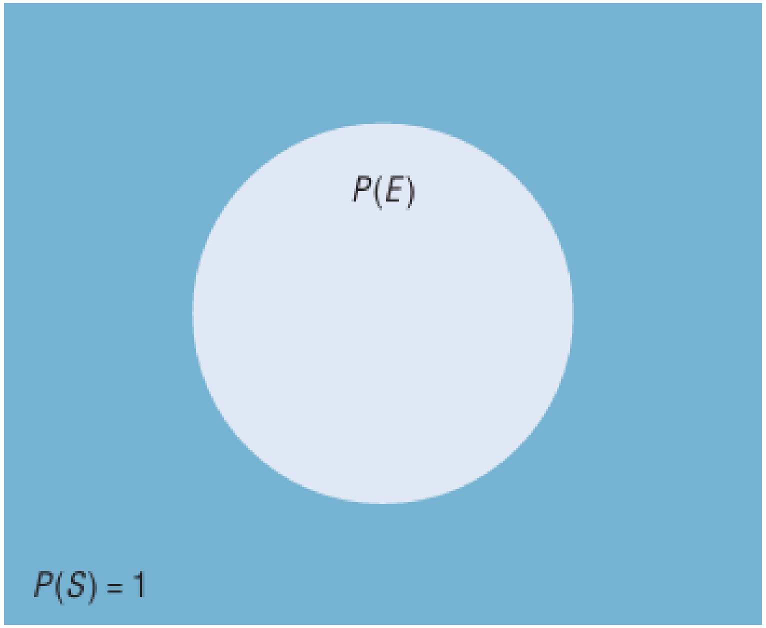
Event A Event B

*"Probability of **event A and event B** equals the probability of **event A** times the probability of **event B given event A**"*

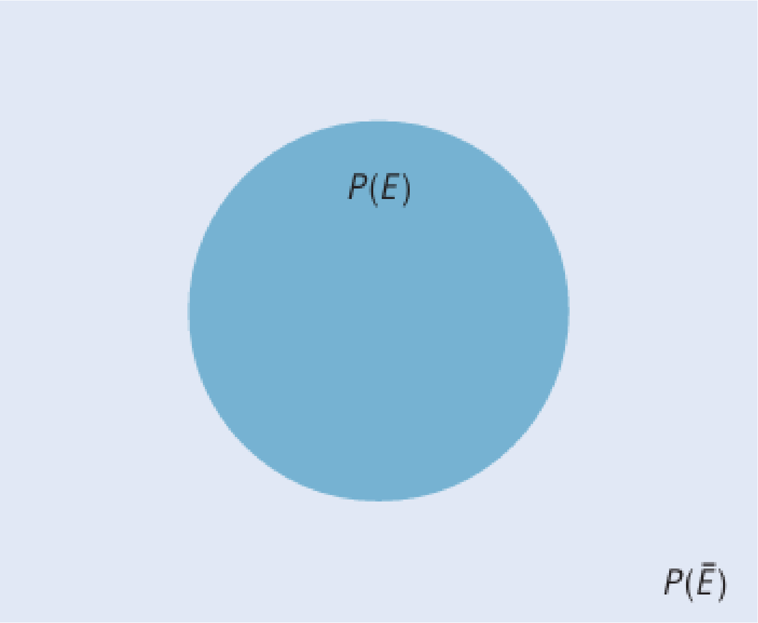
# Venn Diagram




**(a)** Simple probability **(b)** $P(\bar{E}) = 1 - P(E)$




**(a)** Simple probability




**(b)** $P(\bar{E}) = 1 - P(E)$

## EXAMPLE 4–11 Favorite Ice Cream Flavors

In a study, it was found that 23% of the people surveyed said that vanilla was their favorite flavor of ice cream. If a person is selected at random, find the probability that the person's favorite flavor of ice cream is not vanilla.

Source: Rasmussen Report.

**SOLUTION**

$$\begin{aligned} P(\text{not vanilla}) &= 1 - P(\text{vanilla}) \\ &= 1 - 0.23 = 0.77 = 77\% \end{aligned}$$

# Assessing Probability

1. Theoretical/Classical probability—based on theory (*a priori* understanding of a phenomena)

   e.g.: theoretical probability of rolling a 2 on a standard die is 1/6

   theoretical probability of choosing an ace from a standard deck is 4/52

   theoretical probability of getting heads on a regular coin is 1/2

2. Empirical probability—based on empirical data

   e.g.: you toss an irregular die (probabilities unknown) 100 times and find that you get a 2 twenty-five times; empirical probability of rolling a 2 is 1/4

   empirical probability of an Earthquake in Bay Area by 2032 is .62 (based on historical data)

   empirical probability of a lifetime smoker developing lung cancer is 15 percent (based on empirical data)

# Empirical Probability

| Method | Frequency |
|---|---|
| Drive | 41 |
| Fly | 6 |
| Train or bus | 3 |
| | 50 |

**Formula for Empirical Probability**

Given a frequency distribution, the probability of an event being in a given class is

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

This probability is called *empirical probability* and is based on observation.

## EXAMPLE 4–12 Travel Survey

In the travel survey just described, find the probability that a person will travel by airplane over the Thanksgiving holiday.

**SOLUTION**

$$P(E) = \frac{f}{n} = \frac{6}{50} = \frac{3}{25}$$

*Note:* These figures are based on an AAA survey.

## EXAMPLE 4–13 Distribution of Blood Types

In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood, and 2 had type AB blood. Set up a frequency distribution and find the following probabilities.

*a.* A person has type O blood.

*b.* A person has type A or type B blood.

*c.* A person has neither type A nor type O blood.

*d.* A person does not have type AB blood.

Source: The American Red Cross.

**SOLUTION**

| Type | Frequency |
|---|---|
| A | 22 |
| B | 5 |
| AB | 2 |
| O | 21 |
| | Total 50 |

*a.* $P(\text{O}) = \dfrac{f}{n} = \dfrac{21}{50}$

*b.* $P(\text{A or B}) = \dfrac{22}{50} + \dfrac{5}{50} = \dfrac{27}{50}$

(Add the frequencies of the two classes.)

*c.* $P(\text{neither A nor O}) = \dfrac{5}{50} + \dfrac{2}{50} = \dfrac{7}{50}$

(Neither A nor O means that a person has either type B or type AB blood.)

*d.* $P(\text{not AB}) = 1 - P(\text{AB}) = 1 - \dfrac{2}{50} = \dfrac{48}{50} = \dfrac{24}{25}$

(Find the probability of not AB by subtracting the probability of type AB from 1.)

## EXAMPLE 4–14 Hospital Stays for Knee Replacements

Hospital records indicated that knee replacement patients stayed in the hospital for the number of days shown in the distribution.

| Number of days stayed | Frequency |
|---|---|
| 3 | 15 |
| 4 | 32 |
| 5 | 56 |
| 6 | 19 |
| 7 | 5 |
| | 127 |

Find these probabilities.

*a.* A patient stayed exactly 5 days.

*b.* A patient stayed fewer than 6 days.

*c.* A patient stayed at most 4 days.

*d.* A patient stayed at least 5 days.

**SOLUTION**

*a.* $P(5) = \dfrac{56}{127}$

*b.* $P(\text{fewer than 6 days}) = \dfrac{15}{127} + \dfrac{32}{127} + \dfrac{56}{127} = \dfrac{103}{127}$

(Fewer than 6 days means 3, 4, or 5 days.)

*c.* $P(\text{at most 4 days}) = \dfrac{15}{127} + \dfrac{32}{127} = \dfrac{47}{127}$

(At most 4 days means 3 or 4 days.)

*d.* $P(\text{at least 5 days}) = \dfrac{56}{127} + \dfrac{19}{127} + \dfrac{5}{127} = \dfrac{80}{127}$

(At least 5 days means 5, 6, or 7 days.)

# Important Exercise for Exam

- Exercises 4–1 (From Book)
- Page 223

Two events are **mutually exclusive events** or **disjoint events** if they cannot occur at the same time (i.e., they have no outcomes in common).

In another situation, the events of getting a 4 and getting a 6 when a single card is drawn from a deck are mutually exclusive events, since a single card cannot be both a 4 and a 6. On the other hand, the events of getting a 4 and getting a heart on a single draw are not mutually exclusive, since you can select the 4 of hearts when drawing a single card from an ordinary deck.

## EXAMPLE 4–15 Determining Mutually Exclusive Events

Determine whether the two events are mutually exclusive. Explain your answer.

*a.* Randomly selecting a female student
Randomly selecting a student who is a junior

*b.* Randomly selecting a person with type A blood
Randomly selecting a person with type O blood

*c.* Rolling a die and getting an odd number
Rolling a die and getting a number less than 3

*d.* Randomly selecting a person who is under 21 years of age
Randomly selecting a person who is over 30 years of age

**SOLUTION**

*a.* These events are not mutually exclusive since a student can be both female and a junior.

*b.* These events are mutually exclusive since a person cannot have type A blood and type O blood at the same time.

*c.* These events are not mutually exclusive since the number 1 is both an odd number and a number less than 3.

*d.* These events are mutually exclusive since a person cannot be both under 21 and over 30 years of age at the same time.

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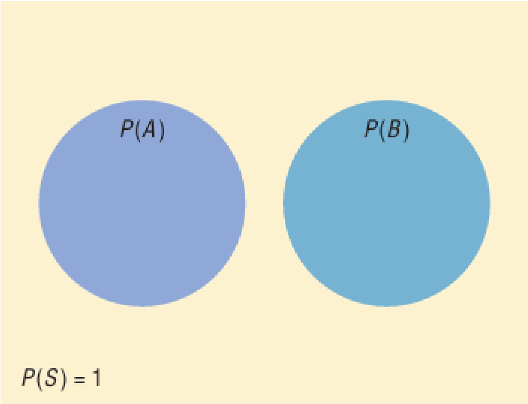



Mutually exclusive events
$P(A \text{ or } B) = P(A) + P(B)$

The probability of two or more events can be determined by the *addition rules.* The first addition rule is used when the events are mutually exclusive.

**Addition Rule 1**

When two events $A$ and $B$ are mutually exclusive, the probability that $A$ or $B$ will occur is

$$P(A \text{ or } B) = P(A) + P(B)$$

Figure 4–6 shows a Venn diagram that represents two mutually exclusive events $A$ and $B$. In this case, $P(A \text{ or } B) = P(A) + P(B)$, since these events are mutually exclusive and do not overlap. In other words, the probability of occurrence of event $A$ or event $B$ is the sum of the areas of the two circles.

## EXAMPLE 4–17 Coffee Shop Selection

A city has 9 coffee shops: 3 Starbucks, 2 Caribou Coffees, and 4 Crazy Mocho Coffees. If a person selects one shop at random to buy a cup of coffee, find the probability that it is either a Starbucks or Crazy Mocho Coffees.

**SOLUTION**

Since there are 3 Starbucks and 4 Crazy Mochos, and a total of 9 coffee shops, $P(\text{Starbucks or Crazy Mocho}) = P(\text{Starbucks}) + P(\text{Crazy Mocho}) = \frac{3}{9} + \frac{4}{9} = \frac{7}{9} \approx 0.778$. The events are mutually exclusive.

## EXAMPLE 4–18 Research and Development Employees

The corporate research and development centers for three local companies have the following numbers of employees:

| | |
|---|---|
| U.S. Steel | 110 |
| Alcoa | 750 |
| Bayer Material Science | 250 |

If a research employee is selected at random, find the probability that the employee is employed by U.S. Steel or Alcoa.

Source: *Pittsburgh Tribune Review.*

**SOLUTION**

$$P(\text{U.S. Steel or Alcoa}) = P(\text{U.S. Steel}) + P(\text{Alcoa})$$

$$= \frac{110}{1110} + \frac{750}{1110} = \frac{860}{1110} = \frac{86}{111}$$

## EXAMPLE 4–19 Favorite Ice Cream

In a survey, 8% of the respondents said that their favorite ice cream flavor is cookies and cream, and 6% like mint chocolate chip. If a person is selected at random, find the probability that her or his favorite ice cream flavor is either cookies and cream or mint chocolate chip.

Source: Rasmussen Report.

**SOLUTION**

$$
\begin{aligned}
&P(\text{cookies and cream or mint chocolate chip})\\
&\quad = P(\text{cookies and cream}) + P(\text{mint chocolate chip})\\
&\quad = 0.08 + 0.06 = 0.14 = 14\%
\end{aligned}
$$

These events are mutually exclusive.

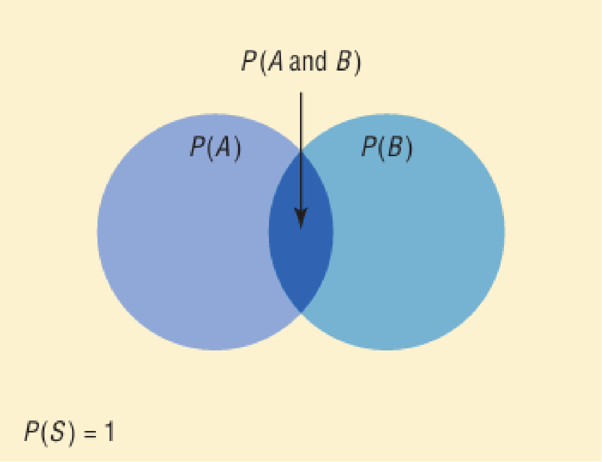
## Addition Rule 2

If $A$ and $B$ are *not* mutually exclusive, then

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

*Note:* This rule can also be used when the events are mutually exclusive, since $P(A \text{ and } B)$ will always equal 0. However, it is important to make a distinction between the two situations.

Figure 4–7 represents the probability of two events that are *not* mutually exclusive. In this case, $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. The area in the intersection or overlapping part of both circles corresponds to $P(A \text{ and } B)$; and when the area of circle $A$ is added to the area of circle $B$, the overlapping part is counted twice. It must therefore be subtracted once to get the correct area or probability.




Nonmutually exclusive events
$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

## EXAMPLE 4–20 Drawing a Card

A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either an ace or a black card.

**SOLUTION**

Since there are 4 aces and 26 black cards (13 spades and 13 clubs), 2 of the aces are black cards, namely, the ace of spades and the ace of clubs. Hence, the probabilities of the two outcomes must be subtracted since they have been counted twice.

$$P(\text{ace or black card}) = P(\text{ace}) + P(\text{black card}) - P(\text{black aces})$$

$$= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13} \approx 0.538$$

## EXAMPLE 4–21 Selecting a Medical Staff Person

In a hospital unit there are 8 nurses and 5 physicians; 7 nurses and 3 physicians are females. If a staff person is selected, find the probability that the subject is a nurse or a male.

**SOLUTION**

The sample space is shown here.

| Staff | Females | Males | Total |
|---|---|---|---|
| Nurses | 7 | 1 | 8 |
| Physicians | 3 | 2 | 5 |
| Total | 10 | 3 | 13 |

The probability is

$$P(\text{nurse or male}) = P(\text{nurse}) + P(\text{male}) - P(\text{male nurse})$$

$$= \frac{8}{13} + \frac{3}{13} - \frac{1}{13} = \frac{10}{13} \approx 0.769$$

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# Combining events (ASW, 157-160)

What is the probability that more than one event has occurred? If there are two events:

Both events could occur – this is referred to as the **intersection** of the two events.

At least one of the events could occur – this is referred to as the **union** of the two events.

Neither event occurs.

# Union of two events

- The **union** of events *A* and *B* is the event containing all the sample points of either *A* or *B*, or both (ASW, 157).
- The notation for the union is $P(A \cup B)$.
- Read this as "probability of *A* union *B*" or the "probability of *A* or *B*."
- The probability of *A* or *B* is the sum of the probabilities of all the sample points that are in either *A* or *B*, making sure that none are counted twice.

# Intersection of two events

- The **intersection** of events *A* and *B* is the event containing only the sample points belonging to both *A* and *B* (ASW, 158).
- The notation for the intersection is P($A \cap B$).
- Read this as "probability of *A* intersection *B*" or the "probability of *A* and *B*."
- The probability of *A* and *B* is the sum of the probabilities of all the sample points common to both *A* and *B*.

# Addition law (ASW, 158)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The probability that at least one event occurs is the probability of one event plus the probability of the other. But to avoid double counting, the probability of the intersection of the two events is subtracted.

# Examples

- Probability of a randomly selected student being an Economics major ($E$) or having high Excel skills ($H$)?

$$\begin{aligned} P(E \cup H) &= P(E) + P(H) - P(E \cap H) \\ &= 24/46 + 6/46 - 2/46 \\ &= 28/46 = 0.609 \end{aligned}$$

- Probability of Business ($B$) or low skills ($L$)?

$$\begin{aligned} P(B \cup L) &= P(B) + P(L) - P(B \cap L) \\ &= 28/46 = 0.609 \end{aligned}$$

- P(Other or medium) = 0.456

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# Mutually exclusive events (ASW, 160)

- Two events are **mutually exclusive** if the events have no sample points in common.
- If two events *A* and *B* are mutually exclusive, the probability of *A* and *B* is zero.
- In this case, the probability of *A* or *B* is the sum of the probability of *A* and the probability of *B*. That is,

  $P(A \cap B) = 0$ if *A* and *B* are mutually exclusive.

  Then $P(A \cup B) = P(A) + P(B)$

# Example of mutually exclusive events

If 3 persons are randomly selected from a large population of half females and half males, what is the probability of selecting 3 females (*A*) and at least 2 males (*B*)?

Event *A* has sample point (F, F, F) .

Event *B* has sample points (F, M, F), (M, F, M), (M, M, F) and (M, M, M) .

Events A and B have no sample points in common, so $P(A \cap B) = 0$.

# Conditional probabilities (ASW, 162-167)

- A **conditional probability** refers to the probability of an event *A* occurring, given that another event *B* has occurred.
- Notation: P(*A* | *B*)
- Read this as the "conditional probability of *A* given *B*" or the "probability of *A* given *B*."
- These are especially useful in economic analysis because probabilities of an event differ, depending on other events occurring.

# Formulae for conditional probabilities

- The probability of *A* given *B* is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- The probability of *B* given *A* is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

# Number of students by major and Excel skill level

| Major of student | Excel skill level | | | | Total |
|---|---|---|---|---|---|
| | None (N) | Low (L) | Medium (M) | High (H) | |
| Math (MA) | 0 | 2 | 4 | 0 | 6 |
| Business (B) | 1 | 3 | 6 | 3 | 13 |
| Economics (E) | 2 | 12 | 8 | 2 | 24 |
| Other (O) | 0 | 1 | 1 | 1 | 3 |
| Total | 3 | 18 | 19 | 6 | 46 |

This table contains the same data as examined earlier, but reorganized as a table rather than in a tree diagram.

# Examples of conditional probabilities from student survey

- Probability that each major has low skill level?

  $P(L \mid MA) = P(L \cap MA) / P(MA) = (2/46) / (6/46) = 2/6 = 0.333$

  $P(L \mid B) = 3 / 13 = 0.231$

  $P(L \mid E) = 0.500$

  $P(L \mid O) = 0.333$

If a student has a high skill level is Excel, what is the probability his or her major is Business? Other?

  $P(B \mid H) = P(B \cap H) / P(H) = (3/46) / (6/46) = 3/6 = 0.500$

  $P(O \mid H) = 0.167$

# Thank You

atanu.Shuvam@ulab.edu.bd

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